

MERN Job Portal - Day 01

Date: 06-07-2026

✓ Work Done

- Created main project folder `JobPortal`.
 - Created two folders:
 - `backend`
 - `frontend` (*not started yet*)
 - Entered the `backend` folder.
 - Installed required npm packages.
 - Created `index.js`.
 - Created Express server.
 - Created `.env` file.
 - Created `utils/db.js`.
 - Connected backend with MongoDB Atlas.
 - Fixed MongoDB DNS connection issue.
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📁 Files Created

```
JobPortal/  
├── backend/  
│   ├── index.js  
│   ├── .env  
│   └── utils/  
│       └── db.js  
└── frontend/
```

📦 Packages Installed

```
npm install express mongoose nodemon jsonwebtoken bcryptjs cookie-parser  
dotenv cors
```

Purpose of Each Package

- **Express** → Creates the backend server and APIs.
- **Mongoose** → Connects Node.js with MongoDB.
- **Nodemon** → Automatically restarts the server after changes.
- **jsonwebtoken** → Will be used later for user authentication.
- **bcryptjs** → Will hash user passwords before storing them.
- **cookie-parser** → Reads cookies from incoming requests.

- **dotenv** → Loads values from the `.env` file.
 - **cors** → Allows the React frontend to communicate with the backend.
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★ Important Concepts

Express Server

Created the first backend server using Express and started it on **Port 8000**.

Environment Variables

Stored the MongoDB connection string and port number inside `.env` to avoid hardcoding sensitive values.

MongoDB Connection

Created a separate `db.js` file to keep the database connection logic separate from the main server file.

CORS

Allowed requests from the React frontend running on `http://localhost:5173`.

✗ Error Faced

MongoDB DNS Error

MongoDB Atlas connection was failing because of a DNS resolution issue.

Solution

Added:

```
import dns from "dns";

dns.setServers(["8.8.8.8", "8.8.4.4"]);
```

After this change, MongoDB connected successfully.

🧠 What I Understood Today

I am still new to backend development, so I don't understand every line of code yet.

However, I understood the overall purpose of today's work:

- Express is used to create the backend server.
- MongoDB stores application data.
- `.env` stores secret configuration values.
- `db.js` is responsible for connecting to MongoDB.
- The backend server starts first, then connects to the database.

Even if I don't understand everything right now, I can understand **why each file is created** and **what its purpose is**.

Next Goal

- Create MongoDB Models.
- Understand how data is stored inside the database.
- Continue backend development.